# Evidence-Grounded Multi-Agent Peer Review: A Reproducible Author–Reviewer–Chair Improvement Loop

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## Abstract

We specify and audit an evidence-grounded, 14-agent paper-review pipeline built around the repository's nineteen context tasks and strict seven-field review form. The implementation separates extraction, specialist critique, synthesis, author rebuttal, and chair output; isolates reviewer and author memories; and keeps decisions out of paper-time requests. A public 20-case ReviewBench–ReviewRebuttal seed drives a leakage-safe, versioned calibration and memory loop. Canonical run 77771e943444d32dc716 verifies execution, convergence, best-state restoration, and one sealed test. It uses deterministic structural signals, not an LLM reviewer, and therefore provides pipeline evidence rather than reviewer-quality evidence.


## 1. Scope and Local Contracts

The system is an offline research instrument for public review histories and author-authorized feedback. It is not permission to delegate a confidential or official ICML review to an LLM. ICML 2026 defines reviewer responsibilities and restricts LLM assistance (International Conference on Machine Learning, 2026d;c); the deployment boundary in Section 5 follows that policy.

Nineteen context tasks. The authoritative prompts.py map has four groups. Main Paper (4) contains Paper Summary, Supplemental Summary, Introduction, and Related Works. Methodology (6) contains Preliminaries, Framework, Training, Proofs, Inference and Application, and Method Summary. Experiments (7) contains Datasets, Implementation Details, Evaluation Metrics, Quantitative Results, Qualitative Results, Ablation Study, and Results Summary. Conclusion (2) contains Limitations and Future Works and Conclusion. Each extraction request must return separate ANSWER and SOURCES sections.

The implemented parser wraps each extraction in an immutable Evidence record with task ID, answer, source strings, provenance, and a content-derived ID. SharedContextStore appends and deduplicates these records and returns frozen ContextPacket snapshots. Sources are still strings: page geometry, typed claims, issues, resolutions, and grounding checks are not implemented.

Strict final form. Only the chair emits ReviewOutput: Soundness, Presentation, Significance, and Originality are integers 1–4; Overall Recommendation is 1–6; Confidence is 1–5; and Comment is nonempty text. The Markdown parser requires the exact seven headings, order, blank-line structure, integer types, and ranges. It does not establish factual accuracy, evidence grounding, or comment quality. This is a repository-specific subset of the full ICML form.

## 2. Team and Information Boundaries

The registry contains exactly 14 agents: five domain specialists (CV, Core ML, NLP, RecSys, General), six criterion specialists (Soundness, Presentation, Significance, Originality, Reproducibility, Ethics), one review synthesizer, one author, and one chair. A deterministic English/Korean keyword router selects at most two domain agents, using General only as fallback.

ReviewerOrchestrator runs all nineteen extraction calls in a bounded pool, restores prompt order, and serializes the resulting packet. Routed domain and all criterion calls then run in parallel. Their critiques and the synthesis are free-form strings: prompts request strengths, weaknesses, questions, and evidence IDs, but no intermediate schema or grounding validator enforces them. The chair receives serialized context

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and synthesis and must pass only the strict final-form parser. With the optional author loop, the author receives context plus the validated initial review and returns a free-form rebuttal; the chair then receives that rebuttal and emits a second validated form. Neither chair call receives a human review or final decision.

The live review –state best_state.json path uses learned state with hard role separation. Lexical cue retrieval first excludes memory from the current forum. Raw training reviews and rebuttals remain internal; memory_guidance converts a match into a paper-agnostic checklist. Domain, criterion, and synthesis requests receive only reviewer guidance, while the author receives only author guidance. Chair requests alone receive rubric weights, calibration scale and bias, and state-integrity metadata as advisory priors. Tests assert self-forum exclusion, raw-text non-disclosure, role separation, and delivery to both chair calls.

The implementation also prepends an untrusted-document rule: paper and extracted text cannot override system instructions or reveal unavailable labels. It does not parse PDFs, detect hidden text, perform OCR comparison, retrieve semantic evidence, validate free-form critique provenance, or build a typed issue graph. Those remain required extensions, not reported capabilities.

## 3. Public Data and Versioned Update

OpenReview connects public submissions, reviews, responses, and decisions by forum (OpenReview, 2025a;b). The canonical smoke does not claim a fresh API snapshot. scripts/build_real_seed.py joins public paper Markdown from ReviewBench (CC-BY-4.0) with reviews and rebuttal dialogues from ReviewRebuttal (Apache-2.0) (ReviewBench Dataset Maintainers, 2026; ReviewRebuttal Dataset Maintainers, 2026). It removes author identity fields and keeps cases with a paper, scored review, rebuttal, and decision. data/real/seed_manifest.json records source URLs, source hashes, licenses, selected forum IDs, and output hash. The corpus SHA-256 begins 02586f1cfef1.

The seed has 20 papers, 77 reviews, and 31 rebuttal dialogues: five ICLR 2022, seven ICLR 2023, and eight NeurIPS 2022, with 15 accept and five reject decisions. There is no complete RecSys case, and ReviewBench text may be a revised public version. The implemented SHA-256 forum-ID split is deterministic and disjoint: 16 train, two development, and two sealed test forums under seed ralphton-icml-real-seed-v1. It does not yet group PDF/title near-duplicates or authors. Development and test IDs cannot enter either memory; the test fingerprint is checked and evaluated once after model selection.

LearningState versions four rubric weights, linear overall-score scale and bias, prompt-manifest digest, bounded reviewer/author memories, and parent digest. It neither fine-tunes model weights nor rewrites prompts. On train data, propose_update first estimates score calibration. For available final decisions it computes acceptance probabilities from calibrated overall scores and applies the clipped residual step

$$b \leftarrow b + \eta\lambda_{\mathrm{dec}}\ \mathrm{mean}(y_i - \hat{p}_i), \qquad \lambda_{\mathrm{dec}} = 0.25. \quad (1)$$

The decision is offline train supervision, never a chair input. A candidate is accepted only if development field coverage and complete-output coverage do not regress beyond tolerance and normalized MAE and Brier do not increase beyond tolerance. Thus decision calibration cannot bypass the per-metric gate.

For version $t$, utility $U_t$ averages available field coverage, complete coverage, one minus normalized MAE, and one minus Brier score (Brier, 1950). Behavioral delta $D_t$ measures normalized score/probability change and Comment token-set distance. State delta $R_t$ averages rubric, scale, bias, prompt digest, and memory change. A plateau satisfies

$$|U_t - U_{t-1}| \leq \epsilon_U, \quad D_t \leq \epsilon_D, \quad R_t \leq \epsilon_R. \quad (2)$$

After the minimum iteration, $p$ consecutive plateaus stop training; every accepted state competes with the best non-regressing state, which is restored before the single sealed-test call.

## 4. Canonical Smoke Result

Run 77771e943444d32dc716 identifies eight substantive artifacts in artifacts/real_seed_v1. For fixed inputs and configuration their hashes and run ID are deterministic; run_manifest.json.created_at is execution provenance and changes on rerun. The full-file dev_review_preview.md is wrapper-free Markdown that round-trips through the strict seven-field parser. Prompt-manifest and restored-state SHA-256 values are, respectively, c0562f49640cbe6b33e59a6bcb2e673a 2bd7f04cb6915a95f9805a7cf18bcbc2 and de6d0295 dbd26723e1e6e579824362ec9dab56f1a43eb8277c8ffcea 857330e7.

The registered configuration used minimum/maximum iterations 3/40, patience 3, $(\epsilon_U, \epsilon_D, \epsilon_R) = (0.001, 0.002, 0.002)$, non-regression tolerance 0.02, learning rate 0.1, decision-calibration weight 0.25, and memory limit 64. It converged after five iterations and restored version 1 from best iteration 1, with

Table 1. Canonical deterministic smoke. Coverage and complete-output coverage are 1.0000 throughout; development and test each contain two forums.

| Stage | MAE | Brier | Utility |
|---|---|---|---|
| Baseline dev | 0.7083 | 0.2863 | 0.8762 |
| Best-state dev | 0.7083 | 0.2796 | 0.8778 |
| Sealed test | 0.3750 | 0.1229 | 0.9426 |

64 reviewer and 53 author memory items from train forums. Four later candidates were rejected; the final three zero-change iterations satisfied patience.

Development utility rose by 0.001656 solely because Brier improved; MAE was unchanged. The sealed test has no pre-update counterpart by design. With two development and two test forums, neither difference supports confidence intervals, domain comparison, or generalization. No LLM review, extraction quality study, human annotation, agent ablation, cost study, or robustness benchmark was run; those results remain N/A (not run).

## 5. Policy, Limitations, and Reproducibility

The system must not ingest confidential submissions or private reviewer discussion, and its output must not be submitted as an official ICML review. ICML's LLM and peer-review ethics policies require venue authorization, confidentiality, integrity, and human accountability (International Conference on Machine Learning, 2026c;b). The current no-auth API client and public seed reduce access risk but do not establish blanket reuse rights. Historical decisions encode capacity, fashion, disagreement, and social bias; they are weak calibration supervision, not research truth.

The current evidence core lacks source spans and issue resolution; specialist, synthesis, and rebuttal outputs are unvalidated strings; no PDF security or near-duplicate document pipeline exists; routing is lexical; RecSys is absent; and hosted models may have pretraining contamination or version drift. Human comment evaluation is costly and subjective. Any quality claim therefore needs a larger lawful corpus, document-level deduplication, registered LLM baselines and ablations, blinded domain experts, agreement statistics, robustness tests, cost reporting, and manifest-linked metric inputs. Formatting follows the official ICML 2026 author style (International Conference on Machine Learning, 2026a); it does not imply submission, acceptance, or endorsement.

Conclusion. The repository now provides a testable separation of paper-time evidence, role-specific guidance, offline supervision, calibrated update, convergence, and sealed evaluation. The canonical artifact establishes that these plumbing and integrity contracts execute; reviewer competence remains unmeasured.

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